

Part II: Universal Measurable Estimators for ADAPT-VQE Gradient Selection

Steps 0–4 — methods, results and reproduction

Part II working note

September 2026

Abstract

We implemented Steps 0–4 of the Part II plan on the nine Part I benchmark cases.

- **Step 0.** Each measurement context in fact reads out a maximal group of 2^n commuting Pauli products. We built the tools to exploit this: measurement circuits, the moments of every group from one transform, and exact shot-level sampling. All are validated against dense linear algebra and against Part I.
- **Step 1.** The II-0 baseline runs Part I’s estimator as a shot-level experiment, with pairwise elimination, radius schedule $\gamma = 0.9$ and top-up allocation. It costs 65–76% fewer context-shots than Part I’s M3 as run, with no wrong selections in 1,100 trials.
- **Step 2.** 58–75% of the Pauli products in $\mathcal{B}_0$ are measured by at least two contexts, so coefficient splitting is possible everywhere.
- **Step 3.** With exact covariances, splitting (II-A) lowers the M3 cost by 38–80% across all cases. Ghost Pauli products (II-B) add up to 10 points. Redesign at every elimination breakpoint (oracle II-D) adds up to 19 points, but only under top-up accounting.
- **Prior transfer.** A design fitted to Hartree–Fock covariances and applied to the correlated state is 66–69% *more* expensive than II-0.
- **Step 4.** The design is learned from the shots, with cross-fitting and a guard, and the confidence radii are estimated too, so the run is fully non-oracle. On both H_4 CISD rows II-A then costs 35–41% less than II-0 run the same way, with no wrong selections in 400 trials. The online oracle ceiling is 43–50%. The HF prior on its own is unreliable (69–85% correct), and it helps only when given little weight.

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1 Problem and conventions

At one ADAPT-VQE step, with state $|\psi\rangle$ and pool $\{G_i\}_{i=1}^N$, the task is to identify $i^* = \arg \max_i |g_i|$, where

$$g_i = \langle C_i \rangle_\psi, \quad C_i = [H, G_i] = \sum_\ell A_{i\ell} P_\ell,$$

using as few *context-shots* as possible. One context-shot is one state preparation plus one measurement in one fully commuting (FC) context.

Cases. The nine Part I cases, rebuilt with the current Part I pipeline:

- systems: H₄ square with side 1.0 and 2.0 Å; LiH at $R = 3.0$ Å; H₂O at $R_{\text{OH}} = 0.9572$ and 1.75 Å;
- states: HF and CISD (LiH HF only);
- STO-3G, Jordan–Wigner, spin-orbital UCCSD pool ($N = 26, 92, 140$ on 8, 12, 14 qubits);
- Part I’s validation gate, and Part I’s deterministic first-fit FC partition of $\mathcal{B}_0 = \bigcup_i \text{supp}(C_i)$ into parent contexts $\mathcal{M}_\alpha$.

The rebuilt $|g_i|$ agree with Part I’s committed values to 3×10^{-10} . Three H₄ rows differ in the sign of some g_i . This is an orbital-phase convention, and every quantity reported here is invariant under it.

Confidence and radii. A confidence radius is $r = z_\delta \epsilon$, with ϵ the one-standard-error target and $z_\delta = \Phi^{-1}(1 - \delta/2N)$, where $\delta = 0.05$ (Bonferroni over the full pool, as in Part I). The gap is $|g_{(1)}| - |g_{(2)}|$, and $\Delta_i = |g_{(1)}| - |g_i|$.

Oracle versus measured. Steps 2–3 are oracle: exact covariances drive the designs and exact gradients fix the elimination trajectory. Their numbers are ceilings. Steps 1 and 4 sample measurement outcomes.

2 Measurement model (Step 0)

Measured groups. A context is measured by a Clifford U_α followed by readout of all n qubits. One shot therefore gives the eigenvalues of all 2^n elements of the maximal abelian group $\mathcal{G}_\alpha = \{\pm U_\alpha^\dagger Z^z U_\alpha\}$. A library Pauli P_ℓ is *measured by* α if $P_\ell \in \mathcal{G}_\alpha$, equivalently if it commutes with all n generators of $\mathcal{G}_\alpha$.

Completion. Let r_α be the rank of the parent members in $\mathbb{F}_2^{2n}$. The generators are completed to n by adding, one at a time, the first candidate that commutes with all current generators and lies outside their span. Two candidate orders are used:

- *canonical*: single-qubit Z_q , then X_q , then Y_q ;
- *mass*: library Paulis in decreasing order of $\sum_i |A_{i\ell}|$.

If a candidate list runs out, a vector of the symplectic complement is used. A seeded *random* order serves as a control. “Span” (no completion) is a diagnostic lower bound on overlap, not a circuit.

Circuit synthesis. The circuit is built from the binary tableau $[X|Z]$ of the n generators in four steps:

1. Row-reduce X , and apply H on the non-pivot qubits, which makes X invertible.
2. Row-reduce to $X = I$. This forces Z to be symmetric.
3. Apply S on every qubit with $Z_{qq} = 1$, and CZ on every pair with $Z_{ab} = 1$.
4. Apply H on all qubits.

Signs are not tracked during synthesis. Instead, every Pauli of interest is propagated through the gate list with exact phases ($P = i^k X^x Z^z$), which gives $U_\alpha P_\ell U_\alpha^\dagger = s_\ell Z^z$ directly.

Moments and sampling. With outcome distribution $p_\alpha(b) = |\langle b|U_\alpha|\psi\rangle|^2$, all group means come from one Walsh–Hadamard transform:

$$E_\alpha[\mathbf{z}] = \sum_b p_\alpha(b)(-1)^{\mathbf{z} \cdot \mathbf{b}}, \quad \mu_\ell = s_\ell E_\alpha[\mathbf{z}_\ell], \quad \text{Cov}(P_l, P_m) = s_l s_m (E_\alpha[\mathbf{z}_l \oplus \mathbf{z}_m] - E_\alpha[\mathbf{z}_l]E_\alpha[\mathbf{z}_m]). \quad (1)$$

The exact p_α gives the oracle moments. A histogram of m sampled outcomes gives the empirical ones, with the factor $m/(m-1)$ on the covariance. Sampling m shots is a single multinomial draw over 2^n outcomes, whose cost does not depend on m .

3 Estimators and levels

For a fixed context library, every linear unbiased estimator has the form

$$\hat{g}_i = \sum_\alpha \mathbf{x}_{i\alpha}^\top \hat{\mu}_\alpha, \quad \sum_{\alpha: P_\ell \in \mathcal{G}_\alpha} x_{i\alpha\ell} = A_{i\ell} \quad \forall \ell. \quad (2)$$

Let $M_q = F_{i\alpha} = \sum_\ell x_{i\alpha\ell} P_\ell$ be the fragments, let C hold their coefficients, and let B select each generator’s fragments. Then Eq. (2) is exactly $A = BC$. Every design is checked for $\max |A - BC| \leq 10^{-9} \max |A|$ and saved as the pair (B, C) .

The levels differ only in which coordinates (α, ℓ) may be non-zero:

- II-0** $\ell \in \text{supp}(C_i)$ in its Part I parent context only. This is Part I’s estimator; nothing is free.
- II-A** $\ell \in \text{supp}(C_i)$ in every context that measures it (coefficient splitting).
- II-B** II-A, plus Paulis with $A_{i\ell} = 0$ that are measured by at least two contexts already carrying C_i (zero-sum ghosts). Candidates come from the rest of the library, i.e. other generators’ supports. They are capped at 200 per generator, ranked by the single-context control-variate score $\sum_\alpha \text{Cov}(F_{i\alpha}^{(0)}, P_\ell)^2 / \text{Var } P_\ell$, where $F^{(0)}$ is the II-0 fragment.

Coefficients are post-processing. Any design satisfying Eq. (2) can be applied to every shot already taken.

4 Allocation, trajectories and cost metrics

Allocation. With n_α shots per context, $\text{Var } \hat{g}_i = \sum_\alpha q_{i\alpha}/n_\alpha$, where $q_{i\alpha} = \mathbf{x}_{i\alpha}^\top \Sigma_\alpha \mathbf{x}_{i\alpha}$ is the one-shot fragment variance. The allocation solves

$$\min_{\mathbf{n}} \sum_\alpha n_\alpha \quad \text{s.t.} \quad \text{Var } \hat{g}_i \leq \epsilon^2 \quad \forall i \in \mathcal{A}, \quad n_\alpha \geq c_\alpha. \quad (3)$$

- With $c = 0$ this is Part I’s solver, reused unchanged.
- With committed shots $c \neq 0$ (“top-up”) it is solved by multiplicative dual ascent. The minimiser of the Lagrangian is $n_\alpha = \max(c_\alpha, \sqrt{\sum_i \lambda_i q_{i\alpha}})$, updated by $\lambda_i \leftarrow \lambda_i (\text{Var } \hat{g}_i / \epsilon^2)^{1/2}$. A final one-dimensional search scales the result to the smallest feasible allocation.

Cost metrics. All three reuse Part I’s trajectories.

M1 One allocation over all N arms at $r = \text{gap}/2$.

M3 bound Part I’s exact-limit M3. The breakpoints are the distinct Δ_i ; at each, arms with $\Delta_i \leq 2r$ are active and resolved to the common radius $r = \Delta/2$.

M3 actual Part I’s event-driven M3. An arm is eliminated when $|g_j| - z\sqrt{V_j} > |g_i| + z\sqrt{V_i}$ on its *actual* variance, with each event found by bisection. This is the primary metric.

Accounting over breakpoints.

- *Maxima* (Part I): each breakpoint is solved from scratch, and each context is charged its largest request.
- *Top-up*: each breakpoint solves Eq. (3) with c equal to the shots already committed.

Designs along a trajectory.

- *Static*: one design, optimised for the full pool, is kept throughout.
- *Redesign* (oracle II-D): the design is re-optimised for the active set at every breakpoint.

5 Joint design solver (Step 3)

Problem. For an active set $\mathcal{A}$ the design problem is

$$\min_{\mathbf{x}, \mathbf{n}} \sum_{\alpha} n_{\alpha} \quad \text{s.t.} \quad \sum_{\alpha} q_{i\alpha}(\mathbf{x})/n_{\alpha} \leq \epsilon^2 \quad (i \in \mathcal{A}), \quad \text{Eq. (2)}.$$

It is jointly convex, since each term is quadratic-over-linear. Because it is scale-free, the optimal $\mathbf{x}$ does not depend on ϵ .

Why not alternate. One could alternate exact per-generator designs at fixed $\mathbf{n}$ (a linear solve over the free directions) with Part I's allocation. This is iterative coefficient splitting (ICS), and it is monotone but stalls. It is iteratively reweighted least squares on a sum of norms: once a context reaches $n_{\alpha} = 0$, no coefficient can return to it. On the winner of H₄ side 1.0 HF, ICS ends at 2.590 after 2,000 iterations, against an optimum of 2.3766 (in units of ϵ^{-2}).

Saddle form. Eliminating $\mathbf{n}$ gives

$$\min_{\mathbf{x}} \max_{\lambda \geq 0} \sum_{\alpha} h(Q_{\alpha}) - \epsilon^2 \sum_i \lambda_i, \quad Q_{\alpha} = \sum_i \lambda_i q_{i\alpha}, \quad h(Q) = n + Q/n, \quad n = \max(c_{\alpha}, \sqrt{Q}).$$

It is solved with two nested loops:

- *Inner:* L-BFGS (memory 10, at most 300 iterations) over the free coefficients of all active arms at once. The objective is smoothed by $Q \rightarrow Q + \delta^2$, with δ reduced by a factor 0.3 per outer iteration, from 0.3 to 10^{-7} of the typical context norm.
- *Outer:* 20 multiplicative dual steps per iteration. The loop stops when the exact cost has not improved by 10^{-7} (relative) for 8 iterations.

The reported cost is always the exact allocation of the returned design, so it is feasible whether or not the solver has converged.

Validation.

- The single-generator optimum matches an independent smoothed L-BFGS minimisation to 10^{-4} .
- Small joint instances match SLSQP to 2×10^{-4} .
- The analytic two-context ghost example is reproduced to 10^{-9} .
- Optimised fragment variances match a state-vector computation to 10^{-10} .

On the large problems the solver's optimality is not certified.

6 Online experiment (Steps 1 and 4)

Loop. This is Part I's successive elimination, starting at $r = \max_i |g_i|$ with $r \leftarrow \gamma r$ each round and stopping when one arm remains or $r < 10^{-3} \max_i |g_i|$. Each round does the following:

1. Allocate shots for the active arms at $\epsilon = r/z_{\delta}$ (maxima or top-up), rounded up to integers.
2. Draw one multinomial histogram per context for the new shots.
3. Estimate $\hat{g}$ from the sample means.
4. Eliminate arms, by one of two rules:
 - *marginal:* Part I's Eq. (4);
 - *pairwise:* i is dropped if some j has $|\hat{g}_j| - |\hat{g}_i| > z_{\delta} \sqrt{V_{ij}}$, where $V_{ij} = \text{Var}(\hat{g}_i - s_{ij} \hat{g}_j)$ and $s_{ij} = \text{sign}(\hat{g}_i \hat{g}_j)$.

Every context that carries a fragment receives at least one shot, because deterministic outcomes still have to be observed once. In Step 1 the radii use the oracle fragment (co)variances, as Part I does.

Learned designs (Step 4). The loop above is run with II-A coefficients learned from the data, as follows.

- *Covariance model.* Per context and fold, $\tilde{\Sigma} = \lambda \Sigma^{\text{prior}} + (1 - \lambda) \hat{\Sigma}^{\text{emp}}$ with $\lambda = \nu/(\nu + m)$, where m is the number of shots. $\hat{\Sigma}^{\text{emp}}$ is the unbiased sample covariance of Eq. (1). It is positive semidefinite, and so is every shrinkage of it, so no projection is needed. The priors are the HF determinant (in the CISD problem's own orbital basis), none ($\nu = 0$), or the oracle (a diagnostic).

- *Cross-fitting.* Every context’s shots are split into two folds. The design fitted on fold A (an exact per-generator solve at fold A’s shots) is applied to fold B’s means, and vice versa. The two estimates are averaged. Each half-estimate is unbiased whatever its design.
- *Guard.* A fitted design is kept for generator i only if the other fold’s covariance predicts a lower variance than the II-0 coefficients do. Otherwise i falls back to II-0.
- *Schedule.* The first round is an II-0 pilot. Designs are refitted whenever the shots held have grown by a factor of 1.5.
- *Allocation.* Shots are planned with the fold-A design and the pooled covariance, and topped up from the shots already held.
- *Radii.* Either estimated from the pooled covariance model (fully non-oracle) or oracle (to isolate the design effect). For the cross-fitted estimate, $\text{Cov}(\hat{g}_i, \hat{g}_j) = \frac{1}{4} \sum_f \sum_\alpha \mathbf{x}_{i\alpha}^{f\top} \Sigma_\alpha \mathbf{x}_{j\alpha}^f / n_\alpha^{\bar{f}}$, where $\bar{f}$ is the fold not used to fit $\mathbf{x}^f$.

7 Results

7.1 Step 0: validation

- Circuits map all 2^n elements of random maximal groups ($n \leq 5$) to $\pm Z^z$, and conjugation phases agree with dense matrices.
- Group means and covariances match Part I’s state-vector evaluator, and II-0 fragment covariances match Part I’s context covariances, to 10^{-12} .

Table 1 checks the sampler against the analytic covariance.

Table 1: Shot-level sampler at scale $\times$ Part I’s M1 allocation; 2,000 replicates on H_4 and 400 on LiH. Columns: largest bias of any $\hat{g}_i$ in standard errors; largest $|\widehat{\text{Var}}/\text{Var} - 1|$ (tolerance $5\sqrt{2/R} = 0.158$ and 0.354); largest correlation error (tolerance $5/\sqrt{R} = 0.112$ and 0.250); two-sample KS p against Part I’s Gaussian surrogate for the winner and for the binding generator. All rows pass. The small p for the H_4 2.0 HF winner (0.003) reflects the lattice of outcomes on a determinant, not a bias.

Case	scale	shots	bias (σ)	var. err.	corr. err.	KS winner	KS binding
H_4 1.0 HF	1	2,244,893	2.56	0.066	0.060	0.79	0.21
H_4 1.0 HF	0.001	2,273	1.81	0.051	0.070	0.35	0.64
H_4 1.0 CISD	1	59,671	1.87	0.078	0.075	0.86	0.86
H_4 1.0 CISD	0.001	88	2.26	0.080	0.064	0.86	0.86
H_4 2.0 HF	1	598,129	1.97	0.049	0.074	0.003	0.37
H_4 2.0 HF	0.001	628	1.91	0.080	0.069	0.03	0.41
H_4 2.0 CISD	1	58,094	1.82	0.109	0.065	0.20	0.35
H_4 2.0 CISD	0.001	86	2.79	0.059	0.074	0.53	0.33
LiH 3.0 HF	1	3,564,398	3.53	0.140	0.190	0.32	0.52

7.2 Step 1: II-0 as a shot-level experiment

Gates.

- *Noiseless.* With every estimate set to the exact gradient and real-valued shot counts, the online loop reproduces Part I’s zero-noise trial exactly: ratio 1.0000000 on all nine case/rule pairs. With integer shots the difference is at most 0.08% on H_4 and 1.8% on LiH, where rounding up drops one arm a round early.
- *Sampled.* At $\gamma = 0.5$ with maxima accounting (250 trials; 100 on LiH), the shot-level runs agree with Part I’s Gaussian surrogate within 1.4 standard errors on every row (Table 2).

Baseline. Table 3 gives the II-0 baseline.

The baseline is 65–76% cheaper than Part I’s M3 as run. Most of that comes from Part I’s own two findings: the pairwise rule and a finer radius schedule.

Table 2: Mean cost of Part I’s surrogate against Part II’s shot-level loop, $\gamma = 0.5$, maxima accounting, with the difference in combined standard errors. Part I’s correlated model is unavailable for LiH (too many generator pairs); its independent-noise marginal run is compared.

Case	marginal rule: Part I / Part II	pairwise rule: Part I / Part II
H ₄ 1.0 HF	1,282,876 / 1,377,896 (+1.3)	697,109 / 670,291 (−1.0)
H ₄ 1.0 CISD	151,337 / 147,834 (−1.4)	89,807 / 92,132 (+0.6)
H ₄ 2.0 HF	534,126 / 528,619 (−0.3)	507,751 / 515,224 (+0.0)
H ₄ 2.0 CISD	102,018 / 97,121 (−1.4)	51,726 / 52,096 (+0.1)
LiH 3.0 HF	880,804 / 872,391 (−0.3)	—

Table 3: II-0 with the pairwise rule, shot-level sampling and oracle radii; 250 trials (LiH 100). The first cost column is Part I’s M3 as Part I ran it (marginal rule, $\gamma = 0.5$, maxima). The **bold** column is the Part II baseline used from here on. Entries are mean $\pm$ s.e.m., with median, 90th percentile and number of wrong selections for the baseline.

Case	Part I M3 as run	$\gamma = 0.5$		$\gamma = 0.9$			
		maxima	top-up	maxima	top-up	median / P90	fail
H ₄ 1.0 HF	1,377,896	670,291	790,598	410,172	337,162 $\pm$ 11,144	300,180 / 570,780	0
H ₄ 1.0 CISD	147,834	92,132	92,686	40,994	40,876 $\pm$ 920	39,275 / 61,958	0
H ₄ 2.0 HF	528,619	515,224 [†]	359,909	1,130,462 [†]	164,348 $\pm$ 6,118	152,312 / 297,895	0
H ₄ 2.0 CISD	97,121	52,096	48,195	24,076	25,000 $\pm$ 662	23,030 / 38,735	0
LiH 3.0 HF	872,391	570,427	566,114	312,640	303,322 $\pm$ 5,460	304,076 / 370,885	0

[†]H₄ 2.0 HF has an exactly degenerate runner-up. Under maxima accounting a few trials never separate the tie and run to the radius floor; the medians are 514,906 and 160,433, with one and two wrong selections. Top-up accounting removes the tail.

Table 4: Multiplicity (*mult.*): mean number of contexts whose measured group contains a Pauli of $\mathcal{B}_0$. *Split.*: share of $\mathcal{B}_0$ with multiplicity ≥ 2 . *Mass*: share of $\sum |A|$ on those Paulis for the pool, the winner and the binding generator (the one that sets M1’s cost). CZ: two-qubit gates per measurement circuit. Aux: mean number of II-B candidates per generator before the cap. Supports do not depend on the state, so HF and CISD differ only through the winner and binding generator.

Case	span		canonical		mass completion						aux
	mult.	split.	mult.	split.	mult.	split.	pool	winner	binding	CZ	
H ₄ 1.0 HF	1.71	52%	1.88	58%	1.95	60%	62%	61%	66%	8.4	386
H ₄ 1.0 CISD	1.71	52%	1.88	58%	1.95	60%	62%	67%	67%	8.4	386
H ₄ 2.0 HF	1.71	52%	1.88	58%	1.95	62%	65%	71%	61%	8.2	379
H ₄ 2.0 CISD	1.71	52%	1.88	58%	1.96	62%	65%	68%	87%	8.2	380
LiH 3.0 HF	2.57	67%	2.73	71%	2.95	76%	84%	87%	89%	23.3	19,348
H ₂ O eq HF/CISD	2.72	67%	2.89	70%	3.10	74%	89%	88%	99%	32.2	50,703
H ₂ O str HF/CISD	2.72	67%	2.88	70%	3.11	75%	88%	87%	100%	32.3	50,527

7.3 Step 2: overlap census

Overlap is abundant, and most of it is present before any completion. The binding generator is 89–100% splittable on LiH and H₂O. Coefficient splitting (II-A) passes the go/no-go test.

7.4 Step 3: oracle ceiling

Table 5 gives every metric in context-shots. Percentages are relative to II-0 on the same metric; redesign is compared with II-0 under top-up accounting.

Observations.

- *Every level passes the 10% gate on every case.* On M3 actual, II-A reduces the cost by 38–54% on H₄, 70–76% on LiH and 64–80% on H₂O.
- *Ghosts add little.* II-B adds 2–10 points over II-A on H₄, and none on LiH with the mass completion. With the canonical completion on LiH it is 12 points worse on M3 actual while better on M1. The static design optimises the full-pool (M1-shape) objective, so downstream M3 costs need not be monotone in the level.

Table 5: Oracle ceilings. *Static* designs are optimised for the full pool; *redesign* re-optimises at every breakpoint (top-up accounting). Redesign was run on H₄ only; for H₂O only II-A with the mass completion was run.

Case	compl.	level	M1	M3 bound (max.)	M3 bound (top-up)	M3 actual	redesign (top-up)
H ₄ 1.0 HF	–	II-0	2,244,862	566,950	566,534	569,020	–
	mass	II-A	719,905	358,865	351,942	352,830 (–38%)	241,899 (–57%)
	mass	II-B	671,127	343,891	338,007	339,202 (–40%)	239,617 (–58%)
	canonical	II-A	780,960	382,898	369,746	375,424 (–34%)	293,701 (–48%)
	canonical	II-B	690,112	369,190	356,615	358,077 (–37%)	290,326 (–49%)
H ₄ 1.0 CISD	–	II-0	59,641	53,752	49,327	49,729	–
	mass	II-A	25,497	24,652	23,589	23,102 (–54%)	23,093 (–53%)
	mass	II-B	23,937	23,090	22,274	22,037 (–56%)	21,828 (–56%)
	canonical	II-A	30,796	29,926	27,130	27,745 (–44%)	26,777 (–46%)
	canonical	II-B	28,427	27,599	25,237	25,555 (–49%)	24,981 (–49%)
H ₄ 2.0 HF	–	II-0	598,102	220,388	220,311	228,733	–
	mass	II-A	240,381	127,470	127,432	136,685 (–40%)	89,779 (–59%)
	mass	II-B	162,830	111,606	111,581	113,507 (–50%)	88,597 (–60%)
	canonical	II-A	263,473	150,010	149,048	157,123 (–31%)	92,049 (–58%)
	canonical	II-B	167,208	119,818	119,131	123,328 (–46%)	90,942 (–59%)
H ₄ 2.0 CISD	–	II-0	58,067	33,385	31,528	31,553	–
	mass	II-A	28,023	18,594	18,145	17,978 (–43%)	15,754 (–50%)
	mass	II-B	22,966	16,276	16,008	15,869 (–50%)	13,579 (–57%)
	canonical	II-A	29,169	18,493	18,034	18,041 (–43%)	15,605 (–51%)
	canonical	II-B	24,778	16,445	16,189	16,191 (–49%)	13,415 (–57%)
LiH 3.0 HF	–	II-0	3,563,904	637,633	586,151	333,272	–
	mass	II-A	581,663	118,386	107,085	98,644 (–70%)	–
	mass	II-B	580,654	117,027	105,489	98,998 (–70%)	–
	canonical	II-A	723,831	146,184	132,268	81,027 (–76%)	–
	canonical	II-B	714,750	141,616	127,460	121,174 (–64%)	–
H ₂ O eq HF	–	II-0	1.72×10^8	2.21×10^7	2.16×10^7	1.32×10^7	–
	mass	II-A	2.71×10^7	4.19×10^6	3.85×10^6	2.69×10^6 (–80%)	–
H ₂ O eq CISD	–	II-0	4.12×10^{11}	4.58×10^{10}	4.10×10^{10}	2.44×10^{10}	–
	mass	II-A	8.50×10^{10}	9.89×10^9	8.51×10^9	6.05×10^9 (–75%)	–
H ₂ O str HF	–	II-0	3.86×10^9	2.02×10^7	1.89×10^7	1.46×10^7	–
	mass	II-A	6.05×10^8	5.85×10^6	5.52×10^6	5.25×10^6 (–64%)	–
H ₂ O str CISD	–	II-0	5.72×10^9	1.71×10^8	1.56×10^8	1.22×10^8	–
	mass	II-A	1.54×10^9	5.35×10^7	4.59×10^7	3.89×10^7 (–68%)	–

- *Completion matters modestly.* On H₄, mass is 0–9 points better than canonical. On LiH, canonical is 5 points better for II-A.
- *Redesign needs top-up accounting.* On the M3 top-up bound, redesign improves the depth-dominated H₄ HF rows from –38%/–42% (static) to –57%/–59%, and the CISD rows by 1–7 points. Under maxima accounting it can be worse than static: on H₄ 1.0 CISD, 31,447 against 24,652.
- *Top-up alone helps.* It lowers the II-0 bound by 0–11%, which closes Part I’s open item “allocate incrementally”.

Prior transfer. At fixed geometry, HF and CISD share one context library. A design fitted on the HF-determinant covariances can therefore be evaluated on the CISD state (Table 6).

Table 6: II-A (mass) designs evaluated on the CISD state, relative to II-0.

Geometry	design fitted on	M1	M3 bound	M3 actual
H ₄ side 1.0	CISD (oracle)	25,497 (–57%)	24,652 (–54%)	23,102 (–54%)
	HF, applied to CISD	93,004 (+56%)	89,355 (+66%)	83,940 (+69%)
H ₄ side 2.0	CISD (oracle)	28,023 (–52%)	18,594 (–44%)	17,978 (–43%)
	HF, applied to CISD	83,159 (+43%)	55,577 (+66%)	52,383 (+66%)

On a determinant, outcomes are either deterministic or perfectly (anti)correlated. A design fitted to that structure relies on cancellations the correlated state does not have.

7.5 Step 4: designs learned from the shots

Setup. Truth is the CISD state of H_4 side 1.0 or 2.0. The classical prior is the HF determinant. II-A uses the mass completion, with the pairwise rule, $\gamma = 0.9$, top-up allocation and shot-level sampling. There are 200 trials per row (seed 2), and every row uses the same loop (Sec. 6). Changes are relative to II-0 with the same radii.

Table 7: Learned designs. Mean $\pm$ s.e.m., median, and number of correct selections out of 200. Estimated radii make a run fully non-oracle; “oracle” rows use exact covariances for the radii, and “II-A oracle” also for the design.

Design	radii	H ₄ side 1.0 CISD			H ₄ side 2.0 CISD		
		mean (change)	median	correct	mean (change)	median	correct
II-0	oracle	41,215 $\pm$ 1,095	38,416	200	24,354 $\pm$ 686	23,054	200
II-A, oracle covariance	oracle	23,450 (−43%)	22,152	200	12,125 (−50%)	11,571	200
II-A, HF prior only	oracle	2.47×10^9	2.50×10^9	192	8.96×10^8	8.04×10^8	183
II-0	estimated	41,940 $\pm$ 1,003	41,790	200	24,591 $\pm$ 672	23,333	200
II-A, HF prior only ($\nu = \infty$)	estimated	439,749 (+949%)	36,143	170	197,780 (+704%)	28,266	138
II-A, HF shrunk, $\nu = 1000$	estimated	28,360 (−32%)	27,343	199	23,233 (−6%)	14,546	193
II-A, HF shrunk, $\nu = 100$	estimated	23,850 (−43%)	23,900	200	14,297 (−42%)	13,528	199
II-A, data only ($\nu = 0$)	estimated	24,688 (−41%)	23,138	200	16,045 (−35%)	14,032	200
II-A, data only, no guard	estimated	25,253 (−40%)	24,000	200	32,265 (+31%)	16,364	197

Findings.

- *Learning from the data recovers most of the ceiling, fully non-oracle.* II-A with covariances estimated from the shots, cross-fitting and the guard costs 41% (side 1.0) and 35% (side 2.0) less than II-0 run the same way. Every one of the 400 trials selected the winner. The online oracle ceiling is 43% and 50%.
- *Estimating variances is nearly free at II-0.* Estimated radii cost 1–2% more than oracle radii, with no wrong selections.
- *The HF prior alone fails twice over.*
 - Its design is wrong for the correlated state, so the true variances are large.
 - Its near-zero variances also corrupt both the radii (69–85% correct) and the allocation. With oracle radii, runs stall until the radius floor and cost 10^8 – 10^9 shots.
- *Shrinkage helps only with little prior weight.* $\nu = 100$ is 2–7 points cheaper than data alone, with one wrong selection in 400 trials. $\nu = 1000$ is worse on cost and gives 8 wrong selections in 400.
- *The guard is necessary.* Without it, the side 2.0 run develops a heavy tail (mean 32,265 against median 16,364) and 3 wrong selections.

Caveats.

- The guard chooses between two unbiased estimators using the second moments of the evaluation fold. This could bias the estimate at second order. A fixed-shot test finds no bias above 4σ over 150 replicates.
- Designs are refitted per generator at fixed shots (Sec. 6), not with the joint solver, so the learned cost is an upper bound on what a jointly learned design would achieve.
- Step 4 was run on the two H_4 CISD rows only.

8 Conclusions and next steps

1. The shot-level II-0 baseline (pairwise rule, $\gamma = 0.9$, top-up) is 65–76% cheaper than Part I’s M3 as run, and it is the reference for every Part II level.
2. Overlap is abundant, and the oracle ceiling of coefficient splitting is large: 38–80% on M3 actual across the cases computed.
3. Ghost Pauli products add little. Per-breakpoint redesign pays on depth-dominated rows, but only with top-up accounting.
4. The Hartree–Fock covariance is harmful as a design prior.

5. Learned from the shots, with cross-fitting and the guard, II-A keeps most of its ceiling without any oracle input: -41% and -35% against II-0 on the H_4 CISD rows, with no wrong selections. The default should be data-driven covariances, with at most a weak prior ($\nu \lesssim 100$). The prior should never be used alone.

Next.

1. *Broaden Step 4.* Run it on LiH (flat or no prior) and on the H_2O CISD rows. Swap the per-generator refit for the joint solver.
2. *Learned redesign (II-D).* Refit for the active set at each elimination, with top-up accounting, on the depth-dominated rows, where the oracle adds up to 19 points.
3. *Correlated trajectory states.* Replace the CISD proxy with saved ADAPT trajectory states (Part I's open item), and study the prior weight ν there.
4. *Price the circuits.* The mass completion uses 8–32 CZ gates per context. Compare against a completion constrained by circuit depth.
5. *Leave II-B, II-C and II-E for later.* Ghost Pauli products added little at the oracle level.

9 Reproduction

Code. `part2_estimator_design/` on branch `part2`. It imports Part I's `src/` directly and builds problems into its own `.cache/`.

Environment. WSL Ubuntu, Python 3.12.3, NumPy 2.5.2, SciPy 1.18.1, PySCF 2.14.0, OpenFermion 1.8.1. NumPy ≥ 2 is required.

Modules (`src/`):

- `symplectic`, `clifford`: Pauli algebra, completion and circuits;
- `contexts`: context library and census;
- `sampler`: moments and sampling;
- `design`: levels and the joint solver;
- `allocation`: top-up allocation;
- `planning`: M1 and M3 trajectories;
- `online`: Step 1 loop;
- `learning`: Step 4.

Commands (from `part2_estimator_design/`):

```
python scripts/build_problems.py                # 9 cases, checked against Part I
python -m pytest tests -q                        # 50 tests
python scripts/step0_validate.py --low-shot 0.001 # H4 sampler gate
python scripts/step0_validate.py --cases LiH_R3p0_HF --replicates 400 \
--out runs/step0_lih
python scripts/step1_online_baseline.py --trials 250 --cases <H4 cases>
python scripts/step1_online_baseline.py --trials 100 --cases LiH_R3p0_HF
python scripts/step1_online_baseline.py --gates-only # exact noiseless gate
python scripts/step2_overlap_census.py           # all nine cases
python scripts/step3_oracle_ceiling.py --cases <H4 cases> --redesign
python scripts/step3_oracle_ceiling.py --cases LiH_R3p0_HF \
--strategies mass canonical
python scripts/step3_oracle_ceiling.py --cases <H2O cases> --strategies mass \
--levels II-0 II-A
python scripts/step3_prior_transfer.py
python scripts/step4_learned_designs.py --trials 200 # H4 CISD rows
python scripts/aggregate_results.py
```

Outputs. Each script writes `runs/<case>/<case>_<step>.csv`, with a `*_meta.json` recording package versions and the git commit. Step 3 designs are saved as `*_B.npz` and `*_C.npz`; these are not tracked and are regenerated by the script.

Seeds. Step 0: 0. Step 1: 1 for Part II and 0 for Part I’s surrogate. Step 4: 2.

Runtimes (one desktop, 4 threads per job):

- tests: about 1.5 min;
- Step 1: minutes per H_4 case, about 20 min for LiH;
- Step 3, H_4 with redesign: 15–20 min per case;
- Step 3, LiH: 5–15 min per design;
- Step 3, H_2O : 12–47 min per design;
- Step 4: 4–8 min per configuration and H_4 case (200 trials). The prior-only configuration with oracle radii takes 20–30 min, because its runs stall until the radius floor.